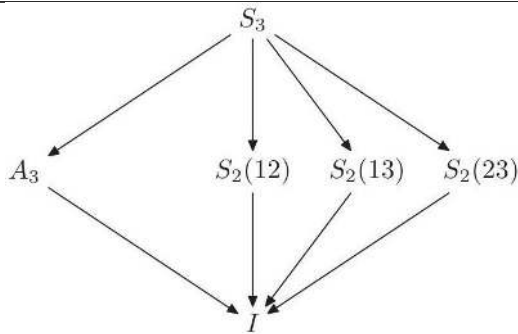
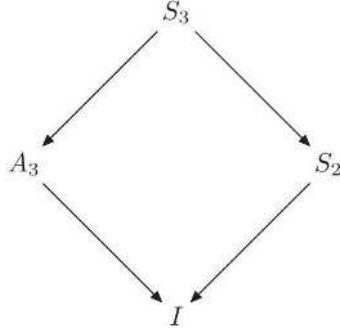
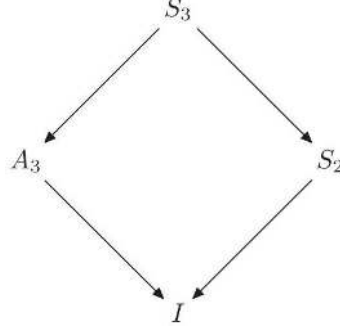
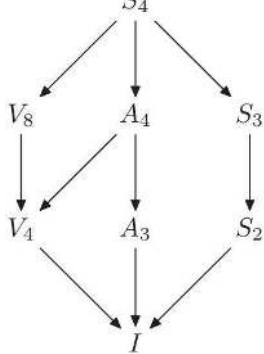
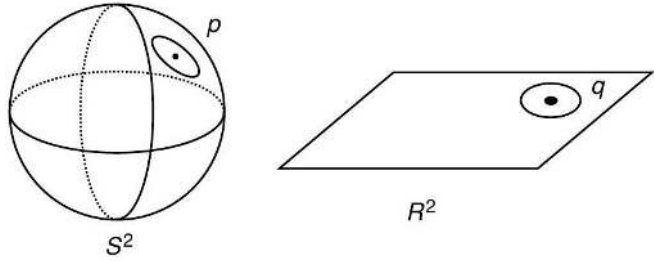
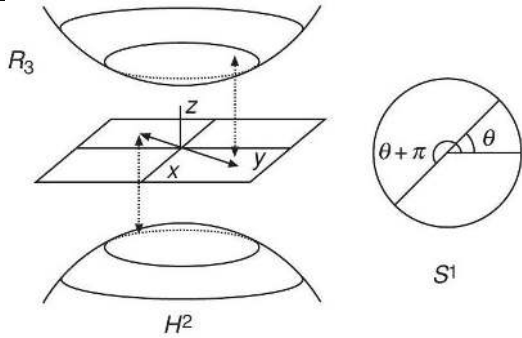
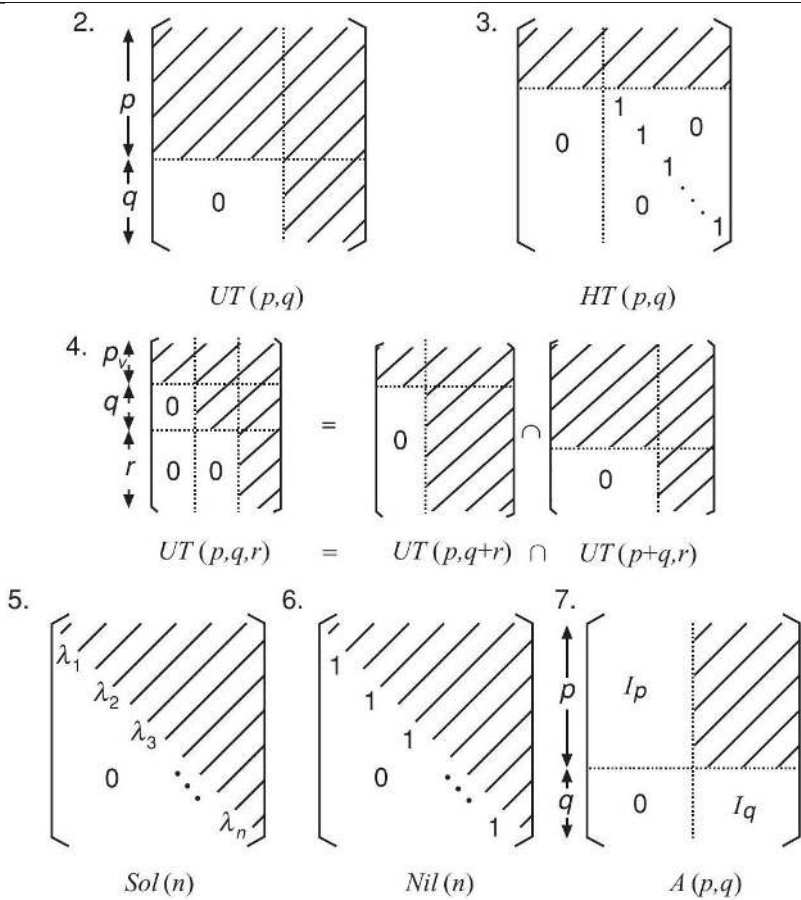
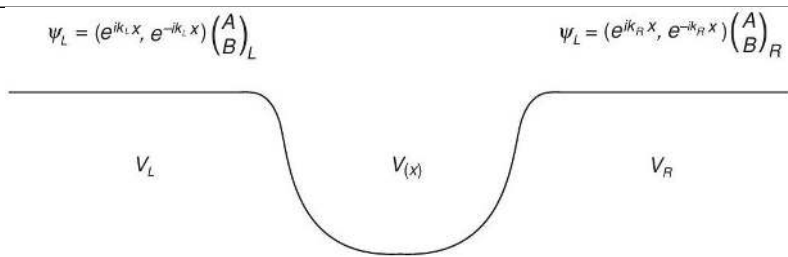


gilmore-lie-groups: image evidence

Image regions

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ DIA_0001	20	—	—	—	
gilmore-lie-groups_ DIA_0002	21	—	—	—	
gilmore-lie-groups_ DIA_0003	24	—	—	—	$S_2 = \{I, (12)\}$ 
gilmore-lie-groups_ DIA_0004	26	—	—	—	
gilmore-lie-groups_ DIA_0005	29	—	—	—	

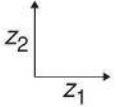
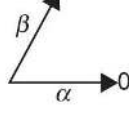
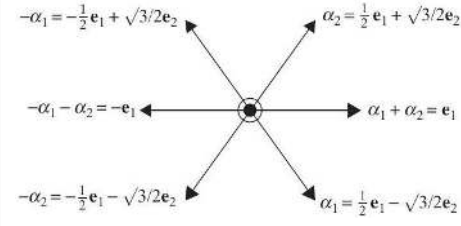
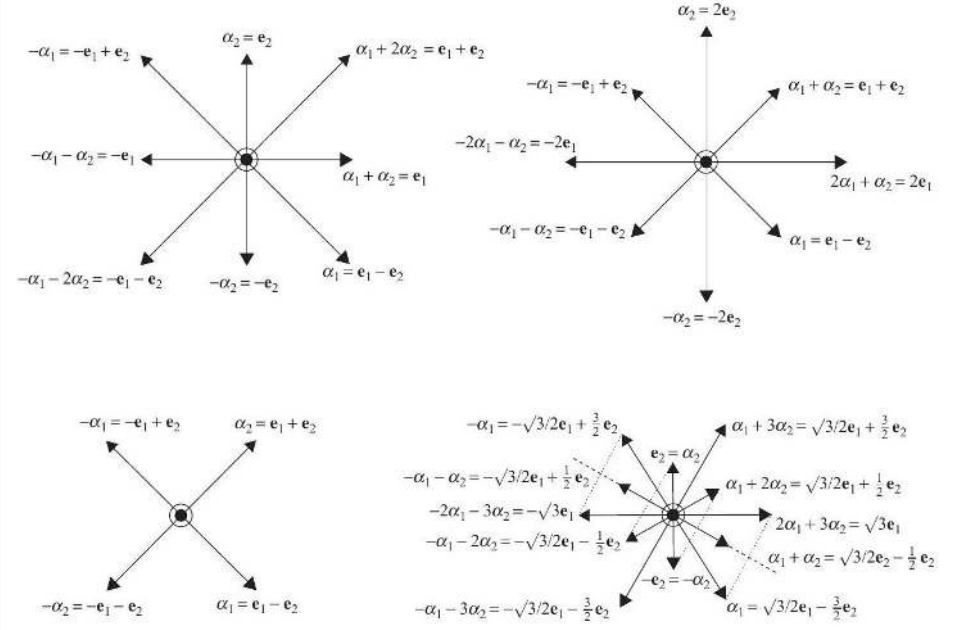
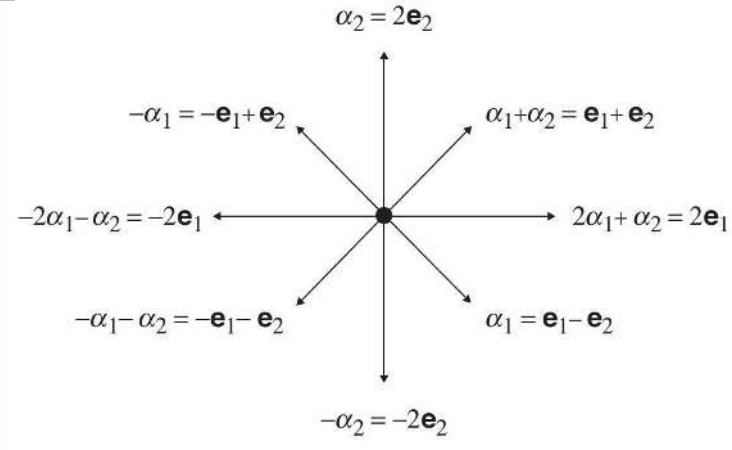
Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ DIA_0006	40	—	—	—	 Diagram illustrating a sphere S^2 with a point p and a plane R^2 with a point q.
gilmore-lie-groups_ DIA_0007	41	—	—	—	 Diagram illustrating a 3D coordinate system with axes x, y, z, a sphere S^1, and a hyperbolic plane H^2.
gilmore-lie-groups_ DIA_0008	51	—	—	—	 Diagram illustrating various matrix structures and partitions, including $UT(p,q)$, $HT(p,q)$, and $Sol(n)$.
gilmore-lie-groups_ DIA_0009	65	—	—	—	 Diagram illustrating a potential energy curve $V(x)$ with regions V_L, $V(x)$, and V_R.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ DIA_0010	67	—	—	—	$\begin{pmatrix} A_L \\ B_L \end{pmatrix} = T_1 \begin{pmatrix} A_M \\ B_M \end{pmatrix} \quad \begin{pmatrix} A_M \\ B_M \end{pmatrix} = T_2 \begin{pmatrix} A_R \\ B_R \end{pmatrix}$ $\begin{pmatrix} A_M \\ B_L \end{pmatrix} = S_1 \begin{pmatrix} A_L \\ B_M \end{pmatrix} \quad \begin{pmatrix} A_R \\ B_M \end{pmatrix} = S_2 \begin{pmatrix} A_M \\ B_R \end{pmatrix}$
gilmore-lie-groups_ DIA_0011	77	—	—	—	nonsemisimple reducible semisimple fully reducible simple irreducible
gilmore-lie-groups_ DIA_0012	89	—	—	—	2. $ut(p, q)$ 3. $ht(p, q)$ 4. $ut(p, q, r) = ut(p, q + r) \cap ut(p + q, r)$ 5. $sol(n)$ 6. $nil(n)$ 7. $a(p, q)$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ DIA_0013	109	—	—	—	
gilmore-lie-groups_ DIA_0014	115	—	—	—	
gilmore-lie-groups_ DIA_0015	117	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ DIA_0016	122	—	—	—	
gilmore-lie-groups_ DIA_0017	136	—	—	—	$ \begin{array}{ccccccc} S & & e^{ikx} & & e^{-\beta\hbar\omega a^\dagger a} & & S^{-1} \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ \begin{bmatrix} 1 & \alpha & \alpha\beta/2 \\ 0 & 1 & \beta \\ 0 & 0 & 1 \end{bmatrix} & & \begin{bmatrix} 1 & ik/\sqrt{2} & -k^2/4 \\ 0 & 1 & ik/\sqrt{2} \\ 0 & 0 & 1 \end{bmatrix} & & \begin{bmatrix} 1 & 0 & 0 \\ 0 & e^{-\beta\hbar\omega} & 0 \\ 0 & 0 & 1 \end{bmatrix} & & \begin{bmatrix} 1 & -\alpha & \alpha\beta/2 \\ 0 & 1 & -\beta \\ 0 & 0 & 1 \end{bmatrix} \end{array} $
gilmore-lie-groups_ DIA_0018	144	—	—	—	1 $\begin{bmatrix} & & \\ & 0 & \\ & & \end{bmatrix}$ zero 2 $\begin{bmatrix} 0 & \text{diag} & \\ & 0 & \text{diag} & \\ & & \ddots & \text{diag} & \\ & & & 0 & \end{bmatrix}$ nil (n) 3 $\begin{bmatrix} \lambda_1 & \text{diag} & \\ & \lambda_2 & \text{diag} & \\ & & \ddots & \text{diag} & \\ & & & \lambda_n & \end{bmatrix}$ sol (n) 4 $\begin{bmatrix} \text{diag} & & \\ & \text{diag} & \\ & & \text{diag} & \end{bmatrix}$ reducible 5 $\begin{bmatrix} \text{diag} & & 0 \\ & \text{diag} & \\ 0 & & \text{diag} \end{bmatrix}$ fully reducible 6 $\begin{bmatrix} \text{diag} & & \\ & \text{diag} & \\ & & \text{diag} \end{bmatrix}$ irreducible
gilmore-lie-groups_ DIA_0019	144	—	—	—	$ \left[\begin{array}{c c} 0 & A \\ \hline 0 & 0 \end{array} \right] \quad \begin{array}{c} \uparrow \\ p \\ \downarrow \\ \uparrow \\ q \\ \downarrow \end{array} $
gilmore-lie-groups_ DIA_0020	148	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ DIA_0021	148	—	—	—	$\left[\begin{array}{c c} * & * \\ \hline 0 & * \end{array} \right] \begin{array}{c} \uparrow V_1 \\ \downarrow \\ \uparrow V_2 \\ \downarrow \end{array} \Rightarrow [\text{Any}, V_2] \subset V_2$
gilmore-lie-groups_ DIA_0022	148	—	—	—	$\left[\begin{array}{c c} * & 0 \\ \hline 0 & * \end{array} \right] \begin{array}{c} \uparrow V_1 \\ \downarrow \\ \uparrow V_2 \\ \downarrow \end{array} \Rightarrow \begin{array}{l} [V_1, V_1] \subseteq V_1 \\ [V_2, V_2] \subseteq V_2 \\ [V_1, V_2] = 0 \end{array}$
gilmore-lie-groups_ DIA_0023	158	—	—	—	$\phi_j(a^r_s) = \frac{1}{j!(n-j)!}$
gilmore-lie-groups_ DIA_0024	158	—	—	—	$\phi_j(X^r_s) = \frac{1}{j!(n-j)!}$
gilmore-lie-groups_ DIA_0025	159	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ DIA_0026	161	—	—	—	$\begin{array}{ccccc} -2 & -1 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ a^2 & a & n+\frac{1}{2} & a^\dagger & a^{\dagger 2} \end{array}$ 
gilmore-lie-groups_ DIA_0027	164	—	—	—	$\begin{array}{ccccccc} \beta-m\alpha & \dots & \beta & \beta+\alpha & \dots & \beta+n\alpha \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}$ 
gilmore-lie-groups_ DIA_0028	165	—	—	—	
gilmore-lie-groups_ DIA_0029	165	—	—	—	
gilmore-lie-groups_ DIA_0030	166	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ DIA_0031	176	—	—	—	
gilmore-lie-groups_ DIA_0032	177	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ DIA_0033	178	—	—	—	
gilmore-lie-groups_ DIA_0034	179	—	—	—	
gilmore-lie-groups_ DIA_0035	180	—	—	—	
gilmore-lie-groups_ DIA_0036	181	—	—	—	
gilmore-lie-groups_ DIA_0037	181	—	—	—	
gilmore-lie-groups_ DIA_0038	181	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ DIA_0039	183	—	—	—	$\begin{array}{c} \alpha_1 \text{---} \alpha_2 \text{---} \cdots \text{---} \alpha_{n-1} \text{---} \alpha_n \\ \alpha_1 \quad \alpha_2 \qquad \qquad \alpha_{n-1} \quad \alpha_n \end{array} \quad A_n$$\begin{array}{c} \alpha_1 \text{---} \alpha_2 \text{---} \cdots \text{---} \alpha_{n-2} \text{---} \alpha_{n-1} \text{---} \alpha_n \\ \alpha_1 \quad \alpha_2 \qquad \qquad \alpha_{n-2} \quad \alpha_{n-1} \quad \alpha_n \end{array} \quad D_n$$\begin{array}{c} \alpha_1 \text{---} \alpha_2 \text{---} \cdots \text{---} \alpha_{n-1} \text{---} \alpha_n \\ \alpha_1 \quad \alpha_2 \qquad \qquad \alpha_{n-1} \quad \alpha_n \end{array} \quad B_n$$\begin{array}{c} \alpha_1 \text{---} \alpha_2 \text{---} \cdots \text{---} \alpha_{n-1} \text{---} \alpha_n \\ \alpha_1 \quad \alpha_2 \qquad \qquad \alpha_{n-1} \quad \alpha_n \end{array} \quad C_n$$\begin{array}{c} \alpha_1 \text{---} \alpha_2 \\ \alpha_1 \quad \alpha_2 \end{array} \quad G_2$$\begin{array}{c} \alpha_1 \text{---} \alpha_2 \text{---} \alpha_3 \text{---} \alpha_4 \\ \alpha_1 \quad \alpha_2 \quad \alpha_3 \quad \alpha_4 \end{array} \quad F_4$$\begin{array}{c} \alpha_6 \\ \\ \alpha_1 \text{---} \alpha_2 \text{---} \alpha_3 \text{---} \alpha_4 \text{---} \alpha_5 \\ \alpha_1 \quad \alpha_2 \quad \alpha_3 \quad \alpha_4 \quad \alpha_5 \end{array} \quad E_6$$\begin{array}{c} \alpha_7 \\ \\ \alpha_1 \text{---} \alpha_2 \text{---} \alpha_3 \text{---} \alpha_4 \text{---} \alpha_5 \text{---} \alpha_6 \\ \alpha_1 \quad \alpha_2 \quad \alpha_3 \quad \alpha_4 \quad \alpha_5 \quad \alpha_6 \end{array} \quad E_7$$\begin{array}{c} \alpha_8 \\ \\ \alpha_1 \text{---} \alpha_2 \text{---} \alpha_3 \text{---} \alpha_4 \text{---} \alpha_5 \text{---} \alpha_6 \text{---} \alpha_7 \\ \alpha_1 \quad \alpha_2 \quad \alpha_3 \quad \alpha_4 \quad \alpha_5 \quad \alpha_6 \quad \alpha_7 \end{array} \quad E_8$
gilmore-lie-groups_ DIA_0040	184	—	—	—	$\begin{array}{c} b_2^+ b_2^+ \\ \uparrow \\ b_1^+ b_2^+ \\ \nearrow \\ b_2^+ b_2^+ + \frac{1}{2} \\ \leftarrow b_1^+ b_1^+ \\ \searrow \\ b_1^+ b_2^+ \\ \downarrow \\ b_2^+ b_2^+ \\ \downarrow \\ b_2^+ b_2^+ \\ \downarrow \\ b_1^+ b_2^+ \\ \nwarrow \\ b_1^+ b_1^+ \\ \swarrow \\ b_1^+ b_2^+ \\ \nearrow \\ b_1^+ b_2^+ \\ \nwarrow \\ b_1^+ b_2^+ \\ \swarrow \\ b_1^+ b_2^+ \\ \nwarrow \\ b_1$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ DIA_0042	200	—	—	—	
gilmore-lie-groups_ DIA_0043	204	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ DIA_0044	222	—	—	—	
gilmore-lie-groups_ DIA_0045	241	—	—	—	Energy eigenvalues, H atom nonrelativistic (darker), relativistic (lighter, deeper)
gilmore-lie-groups_ DIA_0046	249	—	—	—	$\begin{bmatrix} G & \\ & -\frac{1}{2} \end{bmatrix} \begin{matrix} y \\ z_1 \\ z_2 \end{matrix}$ $\begin{bmatrix} G & \\ & -1 \\ & +1 \end{bmatrix} \begin{matrix} y \\ y_{N+1} \\ y_{N+2} \end{matrix}$
gilmore-lie-groups_ DIA_0047	294	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ PIC_0001	250	—	—	—	$-N^3 \times$ Energy of hydrogen atom equal spacing suggests algebra structure
gilmore-lie-groups_ PIC_0002	305	—	—	—	
gilmore-lie-groups_ PIC_0003	308	—	—	—	